

# Canvas Knowledge Graph & Study Assistant

## PRODUCT REQUIREMENTS DOCUMENT

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**Version** 1.0 · **Status** Shipped (v0.2), actively developed · **Last updated** 2026-08-03

**Author** Mihir Argulkar · **Repo** [github.com/mihirargulkar/canvas-obsidian](https://github.com/mihirargulkar/canvas-obsidian) · **License** MIT

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## 1. Summary

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Students pay for Canvas indirectly and for an LLM directly, and the two never meet. Course material sits in Canvas as slide decks, notebooks and announcements that no assistant can read, so students paste screenshots into a chat window and get answers written in a textbook's voice rather than their professor's.

This tool syncs a student's Canvas courses into a local Obsidian vault. Slide decks are transcribed to markdown by a vision model, concepts are extracted and cross linked across lectures, and the whole thing is exposed over MCP so an existing Claude or Gemini subscription can answer questions with citations back to the specific lecture and section.

Everything stays on the student's machine as plain markdown. No hosting, no second subscription, no vendor holding the notes.

**Current state:** v0.2 is running daily against two live courses. 2,077 lines of Python, 64 tests, eight MCP tools, and a working end to end pipeline. Section 5 states exactly which quality claims are measured and which are not.

## 2. Problem

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**Course material is unreadable to the tools students already use.** A lecture is a PPTX. A homework prompt is a PDF linked from an assignment description. A policy change is an announcement. None of it is text an LLM can be pointed at, and none of it is in one place.

**Generic answers are the wrong answers.** A student asking about gradient descent gets the internet's explanation. The exam tests the professor's notation, the professor's emphasis, and the example worked in week four.

The alternatives each give something up:

| OPTION                  | WHAT IT DOES                          | WHAT IT COSTS                                                                         |
|-------------------------|---------------------------------------|---------------------------------------------------------------------------------------|
| Canvas MCP servers      | Broad Canvas API coverage from an LLM | No content understanding; a PPTX stays a PPTX                                         |
| Hosted study assistants | RAG over uploaded files               | Second subscription; course material on someone else's servers; notes you do not keep |
| Manual copy and paste   | Works                                 | Does not scale past one lecture; nothing accumulates                                  |

**The gap:** nothing turns a term of course material into a durable artifact the student owns, queryable by the assistant they already pay for.

### 3. Target user

**Primary.** An undergraduate or graduate student carrying four to six concurrent courses, who already has a Claude subscription or uses the free Gemini CLI, and who will tolerate a one time terminal setup in exchange for not paying again.

**Their week:** four lectures land as slide decks, two homework prompts drop, one announcement moves an exam. By midterms they cannot remember which lecture covered which topic, and searching Canvas returns filenames, not answers.

**Secondary (deferred).** Students who will not open a terminal. See section 12.

**Explicitly not a user: institutions.** Instructure's terms do not permit sharing an access token with a third party. That is fine for a student running this on their own machine and it rules out hosting it for a cohort. This is a product constraint, not an implementation detail, and it shapes every decision below.

### 4. Goals and non-goals

#### Goals

**G1. Make course material answerable.** Any lecture, notebook, homework prompt, announcement or syllabus should be retrievable in natural language, with a citation precise enough to go verify.

**G2. Grounded, not plausible.** Answers come from the student's material or the tool says it does not know. A confident wrong answer about an exam date is worse than no answer.

**G3. Zero marginal cost.** Run on a subscription the student already has. Transcription uses Gemini's free tier. No API bill, no hosting bill.

**G4. The student owns the output.** Plain markdown in a folder, openable in Obsidian, useful even if this tool disappears.

**G5. Stay current without being thought about.** Canvas moves all term. A daily background sync should pick up new material and say what changed.

**Non-goals**

| NON-GOAL                         | WHY                                                                                                                         |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Custom web UI                    | Obsidian already renders wikilinks, backlinks and a graph view. Building that was Phase 5 of the original plan and was cut. |
| Multi-user or hosted SaaS        | Blocked by Instructure’s terms. See section 3.                                                                              |
| Model fine tuning                | Retrieval over ~650 chunks per course does not need it. Evaluated and rejected.                                             |
| Doing the homework               | The tool explains material. What a student does with that is between them and their integrity policy.                       |
| Broad Canvas API coverage        | Grades, quizzes, submissions. Other MCP servers do this well. Content understanding is the differentiator.                  |
| Windows and Linux as first class | Code is cross platform; only macOS is tested and only macOS has the scheduled sync installer.                               |

**5. Success metrics**

Two categories, kept separate on purpose: what has been measured, and what is asserted but not yet instrumented.

## Measured

| METRIC                                | TARGET           | CURRENT                                                          | METHOD                                                          |
|---------------------------------------|------------------|------------------------------------------------------------------|-----------------------------------------------------------------|
| Retrieval recall@5, hand set          | $\geq 0.90$      | <b>1.00</b> (10/10)                                              | Hand labelled gold set,<br><code>tools/eval_retrieval.py</code> |
| Retrieval MRR, hand set               | $\geq 0.85$      | <b>0.90</b> sentence-transformers /<br><b>0.85</b> static + BM25 | Same                                                            |
| Retrieval recall@5, realistic queries | $\geq 0.75$      | <b>0.90</b> (90/100, CI 83-94%)                                  | 100 query synthetic set, pooled relevance labels                |
| Retrieval MRR, realistic queries      | $\geq 0.60$      | <b>0.69</b>                                                      | Same                                                            |
| Recall@5 on the hardest third         | no cliff         | <b>0.88</b> (n=16, low-leakage queries)                          | Same, split by vocabulary overlap                               |
| Index size per course                 | no bloat         | <b>642 chunks</b> (was 1,339)                                    | 51% was base64 image data                                       |
| Generic concepts excluded from graph  | 6/6              | <b>6/6</b>                                                       | <code>tools/eval_graph.py</code>                                |
| Meaningful concept links              | lift over random | <b>61%</b> (67/109) vs <b>8%</b> random, lift +53 pts            | 109 generated pairs,<br><code>tools/make_graph_eval.py</code>   |
| Concept node recall                   | few gaps         | <b>24/109 pairs name a missing node</b>                          | Same; dominant failure mode                                     |
| Fresh install footprint               | < 250 MB         | <b>~170 MB</b> , 74 packages                                     | Down from 1.3 GB / 122 packages                                 |
| Test suite                            | Green in CI      | <b>64 passing</b>                                                | GitHub Actions on every push                                    |
| Re-sync cost when nothing changed     | Zero model calls | <b>Zero</b>                                                      | Content hash cache, verified by run summary                     |

**The hand set is a smoke test, not an instrument.** Ten queries carries a 95% interval of 72 to 100%, and one query moving one rank shifts MRR by 0.10. No fusion weights were ever tuned against it, deliberately.

**Realistic queries score lower than the hand set, and that gap is the point.** 1.00 on the hand set against 0.90 on the synthetic one is not a contradiction: the hand set was written by someone who knew what was in the corpus, so its queries share vocabulary with the notes. Split by vocabulary overlap, the hardest third of the synthetic set still scores 0.88, so there is no cliff. **Treat 0.90 recall and 0.69 MRR as the headline.**

**Two numbers on the way here were wrong, in different ways.** The first, 14%, was arithmetic: the scorer looped over the 10 hand written pairs and divided by the 70 synthetic ones. It was investigated as a retrieval failure and written up as a flaw in exact-source matching, which motivated an LLM judge that then spent three runs fighting a per-day quota. The second, a mean vocabulary overlap of 0.15 cited as evidence the queries were hard, was measured against a corpus half composed of base64 image data; a question invented from gibberish shares no vocabulary with it. On real content it is 0.36.

Both survived because each came with a plausible story. The eval now caches pooled relevance labels in the gold file, so scoring is free, deterministic and needs no judge at measurement time.

**Concept links: 2/6 to 4/6, and the instrument is the limiting factor.** Bisecting the regression showed extraction was never at fault: “Learning Rate” was present in the cached pass 1 output all along, and a prompt edit had dropped it. Three fixes, each defensible on its own terms rather than against the test: pass 2 now adds subsumption edges (“L2 Regularization” to “Regularization”), merges synonym nodes via an `aka` field, and the prompt no longer tells the model to prefer field-standard terms over the note’s own wording, which is what had turned “Basis Function” into “Feature Map”.

The 6 pair test could not separate a real change from run to run variance, so it was replaced by 109 pairs generated from the lecture notes, with the generator never shown the graph. Crucially the harness now also scores 400 **random** concept pairs: “within 2 hops” is trivially won by adding edges, and subsumption linking had just added 25% more of them. The gold pairs sit at 61% against a random baseline of 8%, so the structure is real rather than dense.

The larger set immediately localised the problem. **24 of the 42 failures are pairs naming a concept that was never extracted at all**, against 18 that are genuine missing edges. Node recall, not edge recall, is where the graph loses, which an edge-only metric had been hiding. Owner: section 12, R1.

Asserted, not yet measured

| CLAIM                               | WHY IT IS NOT MEASURED                                                                                                                          | PLAN                                                                         |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Answer quality (not just retrieval) | The judge scores whether a retrieved passage is relevant, not whether the assistant’s final answer is correct                                   | Sample answers, score for grounding and citation accuracy                    |
| A complete judged run               | Both free tier models are capped per day (20/day for <code>gemini-3.5-flash</code> ), and the judge competes with transcription for that budget | Run the 7 judge calls on a day with no sync, or move the judge to a paid key |
| Transcription fidelity              | Spot checked by hand across three decks, never scored                                                                                           | Sample 20 slides, score against source                                       |
| Time saved per student              | One user, no baseline                                                                                                                           | Deferred until there are users to ask                                        |

**A note on measurement discipline.** Three separate bugs in this project shared one shape: *a failure was indistinguishable from a negative result*. The sync reported “no changes” when it had not looked at files. The judge counted quota errors as retrieval misses, twice, once reporting 37% recall that was really an API outage. Two published numbers (14% recall from exact source matching, 37% from the first judged run) were later found invalid and retracted. Any new metric added to this table must distinguish “measured zero” from “did not run”.

6. Requirements

Priority: **P0** ships or the product does not work; **P1** ships in v1.0; **P2** is post v1.0.

## 6.1 Ingestion

| ID | REQUIREMENT                                                                      | PRI | ACCEPTANCE                                                                                |
|----|----------------------------------------------------------------------------------|-----|-------------------------------------------------------------------------------------------|
| I1 | Transcribe PDF, PPTX and DOCX lecture material to markdown                       | P0  | A slide deck produces readable markdown with headings and equations preserved             |
| I2 | Read notebooks, text and markdown without a model call                           | P0  | <code>.ipynb</code> is indexed with zero Gemini calls                                     |
| I3 | Ingest homework: assignment descriptions plus the prompt PDFs linked inside them | P0  | “What is homework 3 asking for” answers from the prompt, not the title                    |
| I4 | Ingest announcements and syllabus                                                | P0  | “Did I miss any announcements” returns posted announcements with dates                    |
| I5 | Never re-transcribe unchanged content                                            | P0  | Second run reports N cached, 0 new                                                        |
| I6 | Survive a rate limit and resume                                                  | P0  | A run that hits 429 leaves prior work cached; the next run continues                      |
| I7 | A cache key change must not strand existing work                                 | P0  | Adding prompt and model to the key migrates old entries rather than re-billing the corpus |
| I8 | Images (PNG, JPG, WEBP) go straight to the vision model                          | P1  | A scanned handout is transcribed                                                          |

## 6.2 Knowledge graph

| ID | REQUIREMENT                                                     | PRI | ACCEPTANCE                                                                          |
|----|-----------------------------------------------------------------|-----|-------------------------------------------------------------------------------------|
| K1 | Extract concepts per lecture with definitions and relationships | P0  | One lecture yields concept notes with YAML frontmatter                              |
| K2 | Merge concepts across lectures by canonical name                | P0  | A concept taught in weeks 2 and 7 is one node linked to both                        |
| K3 | Exclude administrative material from the graph                  | P0  | “Today’s Agenda” and “Course Overview” are not nodes (6/6)                          |
| K4 | Keep homework and notebooks searchable but out of the graph     | P0  | <code>hw-</code> and <code>code-</code> prefixed notes are indexed, never extracted |
| K5 | Output opens in Obsidian with a meaningful local graph          | P0  | Backlinks and graph view work with no plugins                                       |
| K6 | Never delete existing notes before replacements are written     | P0  | A mid write disk error leaves the previous vault intact                             |

### 6.3 Retrieval and answering

| ID | REQUIREMENT                                                                            | PRI | ACCEPTANCE                                                             |
|----|----------------------------------------------------------------------------------------|-----|------------------------------------------------------------------------|
| R1 | Deadline questions answered deterministically from the live API, never from the index  | P0  | “What’s due this week” matches Canvas exactly                          |
| R2 | Backward looking deadline questions                                                    | P0  | “Am I overdue on anything” returns past due items, not a future window |
| R8 | Grades reported, never recomputed                                                      | P0  | Both Canvas totals surfaced; hidden grades return an error, not a zero |
| R3 | Conceptual questions answered from indexed material with lecture and section citations | P0  | Answer names the lecture it came from                                  |
| R4 | Hybrid retrieval: dense embeddings fused with BM25                                     | P0  | Exact term queries and paraphrases both work                           |
| R5 | Say so when material is absent                                                         | P0  | A question about an untaught topic is declined, not invented           |
| R6 | Incremental index updates                                                              | P0  | Editing one note does not rebuild the index                            |
| R7 | Every chunk of every note is indexed                                                   | P0  | A note whose only heading is # still produces chunks                   |

*R7 exists because it once did not hold. Four notes produced zero chunks and were invisible to search. Fixing it recovered 48 chunks and moved MRR from 0.90 to 0.95 on the then current gold set.*

## 6.4 Staying current

| ID | REQUIREMENT                                                                    | PRI | ACCEPTANCE                                                                  |
|----|--------------------------------------------------------------------------------|-----|-----------------------------------------------------------------------------|
| C1 | Report what is new since the last sync                                         | P0  | Output names new announcements, assignments and files                       |
| C2 | A failed read must never be persisted as “seen”                                | P0  | A 403 during sync does not cause the next healthy run to re-report the term |
| C3 | A freshness check must not imply “nothing posted” when files were not examined | P0  | A cheap check still lists untranscribed files by name                       |
| C4 | Scheduled daily sync, quiet unless something changed                           | P1  | launchd job writes to the log only on change                                |
| C5 | Cross platform scheduling                                                      | P2  | Linux systemd timer, Windows Task Scheduler                                 |

*C3 is a shipped bug turned requirement. A fast refresh skipped file listing, reported “no changes”, and a client concluded a lecture that was on Canvas “had not been posted”. Silence about something never checked reads as evidence.*

## 6.5 Multiple classes

| ID | REQUIREMENT                                                    | PRI | ACCEPTANCE                                                                               |
|----|----------------------------------------------------------------|-----|------------------------------------------------------------------------------------------|
| M1 | Detect the current term’s courses without manual configuration | P0  | Correct courses found across schools that never set term end dates                       |
| M2 | Per class isolation: one broken class cannot sink the others   | P0  | A course with a restricted tab is skipped with a warning; the rest still report          |
| M3 | Every tool scopes to one class or spans all of them            | P0  | Optional <code>course</code> slug on all eight MCP tools                                 |
| M4 | One filesystem safe slug definition, used everywhere           | P0  | <code>DS 4400</code> resolves to <code>DS4400</code> in both the vault path and the link |
| M5 | Cross class dashboard                                          | P1  | One file lists deadlines across all classes by date                                      |

*M4 is not cosmetic. Two definitions of “slug” disagreed, so one half of the codebase wrote to `vault/DS4400/` while the other linked `[[DS/Dashboard]]`. A course named `CS1800/1802` also produced nested directories, and a malformed name could produce `../../`.*

## 6.6 Setup and operations

| ID | REQUIREMENT                                                         | PRI | ACCEPTANCE                                                                                |
|----|---------------------------------------------------------------------|-----|-------------------------------------------------------------------------------------------|
| S1 | One command setup                                                   | P0  | <code>./setup.sh</code> creates the venv, installs, prompts for keys, offers a first sync |
| S2 | Re-runnable setup                                                   | P0  | Running it twice is safe                                                                  |
| S3 | Install under 250 MB                                                | P0  | ~170 MB measured                                                                          |
| S4 | Works with Claude Desktop, Claude Code, Gemini CLI, Cursor, VS Code | P0  | One config block documented for each                                                      |
| S5 | Missing credentials fail with instructions, not a traceback         | P0  | Absent <code>.env</code> prints where to get a token                                      |
| S6 | Library code never calls <code>sys.exit</code>                      | P0  | Formatting a date without credentials falls back to the OS timezone                       |

## 6.7 Trust and safety

| ID | REQUIREMENT                                                         | PRI | ACCEPTANCE                                                                                                 |
|----|---------------------------------------------------------------------|-----|------------------------------------------------------------------------------------------------------------|
| T1 | The Canvas token lives only in a gitignored <code>.env</code>       | P0  | Never committed, never passed through MCP                                                                  |
| T2 | All student data is gitignored                                      | P0  | <code>vault/</code> , <code>notes/</code> , <code>cache/</code> , <code>index/</code> never enter the repo |
| T3 | Disclose that course content is sent to Gemini during transcription | P0  | Stated in the README                                                                                       |
| T4 | Academic integrity is the student's call, stated plainly            | P0  | One sentence, no lecture                                                                                   |

## 7. How it works



**Pipeline.** `ingest` downloads files, converts PPTX and DOCX to PDF via LibreOffice, sends vision formats to Gemini and extracts text formats directly. `extract` runs two passes: concepts per lecture, then a merge across lectures that creates the cross lecture links. `chat` chunks notes by section, embeds them, and stores vectors in SQLite. `sync` orchestrates all of it per course and diffs against seen state.

**Retrieval.** Dense cosine similarity over normalised vectors, fused with BM25 by reciprocal rank fusion. Deadline questions bypass all of it and hit the live API, because a stale index answering “when is the midterm” is the one failure mode with real consequences.

**Live corpus (author’s machine, 2026-08-04):** 2 courses, 74 notes, 673 indexed chunks, 178 concept nodes, 294 edges.

### MCP surface

| TOOL                              | PURPOSE                                            | SOURCE OF TRUTH |
|-----------------------------------|----------------------------------------------------|-----------------|
| <code>list_courses</code>         | Current classes with slugs                         | Live API        |
| <code>upcoming_assignments</code> | Deadlines within N days, all classes by default    | Live API        |
| <code>grades</code>               | Current grade per class, plus per-assignment marks | Live API        |
| <code>announcements</code>        | Recent professor updates                           | Live API        |
| <code>syllabus</code>             | Course policies                                    | Live API        |
| <code>search_notes</code>         | Hybrid search over lectures, homework, notebooks   | Local index     |
| <code>concept</code>              | One concept node with its links                    | Local vault     |
| <code>refresh</code>              | What is new since the last sync                    | Both            |

Results are capped at 2,000 characters per chunk. An 800 KB chunk once exceeded the MCP 1 MB message limit and broke the transport.

## 8. Decision log

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**D1. Obsidian instead of a custom web UI.** A React graph view was designed and partly built. Obsidian already ships wikilinks, backlinks, graph view, search and mobile sync, and the output is more valuable as files the student keeps than as a page only this tool can render. *Cost:* users install Obsidian. *Status:* final, and it removed an entire frontend from the project.

**D2. MCP instead of a bundled chat.** Students already pay for Claude or use Gemini free. Calling an API on their behalf adds a bill and a worse model. *Cost:* setup requires editing a client config. *Status:* final.

**D3. SQLite and numpy instead of ChromaDB.** Brute force cosine over a few thousand normalised vectors is a matrix multiply. ChromaDB brought a large dependency tree for an index that does not need one. *Cost:* linear scan. *Ceiling:* fine to roughly 100k chunks, well past a student's term.

**D4. Static embeddings plus BM25 instead of PyTorch.** `sentence-transformers` pulls ~800 MB of PyTorch and scores MRR 0.90 against 0.85 for static embeddings fused with BM25. Default install is the small one; if `sentence-transformers` is present it is picked up automatically. *Cost:* 0.05 MRR by default. *Status:* final, and it is what took the install from 1.3 GB to ~170 MB.

**D5. Gemini for transcription, despite the client already having a model.** Transcription is an unattended batch job over a hundred files. It must run on a schedule at 07:30 with nobody watching, which an interactive MCP client cannot do. *Cost:* one free API key during setup.

**D6. No fine tuning.** Considered and rejected. Retrieval over a term of material is a search problem; a fine tuned model would memorise one student's corpus, need retraining weekly, and lose citations.

**D7. Content hash caching everywhere.** The binding constraint is Gemini's free tier, not compute. Every expensive step is keyed by a hash of its inputs including the prompt and model, so a re-run is free and a prompt change correctly invalidates.

## 9. Risks

| RISK                                                                   | IMPACT                                 | MITIGATION                                                 | STATUS                                            |
|------------------------------------------------------------------------|----------------------------------------|------------------------------------------------------------|---------------------------------------------------|
| Gemini free tier quota during a large first sync                       | First run does not finish              | Everything cached and resumable; partial results usable    | Mitigated                                         |
| Instructure terms forbid token sharing                                 | No hosted version, ever                | Designed as local only from the start                      | Accepted, shapes the product                      |
| Vision transcription errors propagate to concepts and answers          | Wrong answer with a confident citation | Citations point at the source slide so a student can check | Partially mitigated; fidelity unmeasured          |
| Concept extraction quality degrades as the corpus grows                | Graph becomes less useful over a term  | <code>eval_graph.py</code> catches it; it just did, at 2/6 | Detected, unfixed                                 |
| Canvas API changes                                                     | Sync breaks                            | <code>canvasapi</code> is pinned below the next major      | Mitigated                                         |
| Setup friction (terminal, two keys, client config)                     | Users drop off before first value      | <code>setup.sh</code> one shot                             | Partially mitigated; not observed with real users |
| Concurrent write between the daily job and an MCP <code>refresh</code> | Corrupt cache state                    | None. No lockfile.                                         | <b>Open</b>                                       |
| Non-atomic <code>cache/*.json</code> writes                            | Crash mid write corrupts state         | Corrupt state is caught and treated as empty               | Partially mitigated                               |

## 10. Open questions

1. **Can a 6 pair test tell improvement from noise?** No. Extraction is nondeterministic: re-running the same prompt moved one gold pair from a direct link to 3 hops with no code change. Every graph conclusion is currently drawn from six pairs, which is the weakest instrument in the project.
2. **How much of the retrieval score is the synthetic generator's bias?** The queries come from the same model family that answers them, and the difficulty filter is a vocabulary-overlap heuristic. Real student query logs would settle it; there aren't any yet.
3. **Does anyone finish setup?** Two API keys, a terminal, a client config, and optionally LibreOffice. Unknown drop off.
4. **Is one shared vault right for a whole degree?** Currently scoped to the current term. Four years of concepts might be more valuable, or unusable.
5. **Do students want the graph, or just the answers?** The graph is the expensive half of the pipeline and the least measured.

## 11. Dependencies and constraints

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**Runtime:** Python 3.10+, a Canvas access token, a free Gemini API key. LibreOffice is optional and only needed for PPTX and DOCX.

**Direct dependencies (7):** `canvasapi`, `python-dotenv`, `google-genai`, `numpy`, `model2vec`, `mcp[cli]`, `pytest`. Majors are capped deliberately: `mcp` renamed `FastMCP` to `MCPServer` in 2.0, and an unpinned major silently breaks a fresh install.

**Optional:** `sentence-transformers` for +0.05 MRR at ~800 MB.

**Platform:** written cross platform, tested on macOS. The scheduled sync installer is `launchd` only.

## 12. Roadmap

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**R1. Fix concept link recall (next).** Bisect the 5/6 to 2/6 regression. Exit criterion: back to 5/6 with generic exclusion still 6/6.

**R2. Finish labelling, then raise recall.** Two steps, in order.

*First, fix the instrument.* Error analysis on the 28 misses found 23 of them are the same artefact: the query was generated from a notebook, retrieval returned the lecture covering that exact topic, and single-source ground truth called it wrong. `--relabel` now judges retrieved sources once and caches the verdicts in the gold file, so scoring stays free afterwards. 195 pairs, 4 calls. The gold set was also 52/70 notebook-derived because notebooks produce long chunks and the generator filters on length; sampling is now stratified by material type. Regenerate after relabelling.

*Then improve retrieval,* in order of laziness: tune the RRF fusion weights (never tuned, because no gold set was big enough to tune against without overfitting), prepend the lecture and section title to chunk text before embedding, revisit chunk boundaries, then consider a reranker. Exit criterion:  $\text{recall@5} \geq 0.75$  on the relabelled set with the interval clear of the baseline.

Do not tune against the current 0.60. Optimising toward it means teaching the ranker to prefer notebooks over lectures on the same topic, which is not a goal.

**R3. Close the concurrency gap.** A lockfile between the scheduled job and MCP `refresh`, and atomic cache writes.

**R4. Reduce setup friction.** Measure drop off first. Options in order of laziness: better error messages, a `doctor` command, a packaged installer.

**R5. Cross platform scheduling.** `systemd` timer and Task Scheduler.

**Deferred indefinitely:** hosted version (blocked by terms), custom UI (superseded by Obsidian), fine tuning (evaluated and rejected), broad Canvas API coverage (other servers do it).

## Appendix A: Data model

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|                                |                                          |
|--------------------------------|------------------------------------------|
| notes/<CLASS>/Lecture4.md      | transcribed source, YAML frontmatter     |
| notes/<CLASS>/hw-Homework#3.md | homework prompt, excluded from the graph |
| notes/<CLASS>/code-gradient.md | notebook, excluded from the graph        |
| notes/<CLASS>/announcements.md | professor updates                        |
| vault/<CLASS>/concepts/*.md    | one file per concept, [[wikilinks]]      |
| vault/<CLASS>/Dashboard.md     | per class deadlines                      |
| vault/Dashboard.md             | all classes                              |
| index/vectors.db               | SQLite: chunks + embedder id             |
| cache/                         | downloads, transcriptions, seen state    |

Everything above is gitignored. A note is a lecture unless its filename is prefixed `hw-` or `code-`, or it is a poll, solution, announcement or syllabus.

## Appendix B: Glossary

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**Chunk.** A section of a note, split on `##` headings, embedded as one unit.

**Slug.** Filesystem safe short course key, for example `DS4400`. Derived from Canvas `course_code`, falling back to the name. Single definition in `canvas.slug_from`.

**RRF (reciprocal rank fusion).** Combines the dense and BM25 rankings by summing  $1/(k + \text{rank})$ . Something both retrievers like outranks something only one of them found.

**Recipe.** A hash of the prompt plus model, mixed into every cache key so changing either correctly invalidates prior work.

**MRR.** Mean reciprocal rank.  $1/\text{rank}$  of the first correct hit, averaged. Rank 1 scores 1.0, rank 5 scores 0.2.